# Samuel Silverman

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## Research Vision

I develop new approaches to discovering, designing, and manufacturing physical systems. By **integrating artificial intelligence with computational mechanics**, my work builds algorithmic foundations for **making automated physical design fast, reliable, and widely accessible**. With applications spanning diverse engineering domains, this research has the potential to optimize high-performance structures, accelerate the discovery of functional materials, and streamline the path to fabrication.

## Education

*PhD, Computer Science* 2021-Present
Boston University, Boston, MA
Advisor: Emily Whiting

*MS, Computer Science* 2019-2021
Columbia University, New York, NY

*BS, Computer Science* 2015-2019
University of Massachusetts Amherst, Amherst, MA

## Publications

Dylan J. Balter, Colin McMillen, Alec Ewe, Jonathan Thomas, Samuel Silverman, Latitha Parameswaran, Luis Fernando Velásquez-García, Emily Whiting, Steven Patterson, Hilmar Koerner, Keith A. Brown
*Giant Flexoelectricity of Additively Manufactured Polylactic Acid*
**Additive Manufacturing (2026)**

Samuel Silverman, Dylan Balter, Keith A. Brown, Emily Whiting
*Random-Walk Microstructures for Differentiable Topology Optimization*
**ACM Symposium on Computational Fabrication (2025)**

Samuel Silverman, Kelsey L. Snapp, Keith A. Brown, Emily Whiting
*Data-Driven Nonlinear Deformation Design of 3D-Printable Shells*
**3D Printing and Additive Manufacturing (2025)**

Kelsey L. Snapp, Samuel Silverman, Richard Pang, Thomas M. Tiano, Timothy J. Lawton, Emily Whiting, Keith A. Brown
*A Physics-Informed Impact Model Refined by Multi-Fidelity Transfer Learning*
**Extreme Mechanics Letters (2024)**

Kelsey L. Snapp, Benjamin Verdier, Aldair E. Gongora, Samuel Silverman, Adedire D. Adesiji, Elise F. Morgan, Timothy J. Lawton, Emily Whiting, Keith A. Brown
*Superlative Mechanical Energy Absorbing Efficiency Discovered Through Self-Driving Lab-Human Partnership*
**Nature Communications (2024)**

Vishnu Nair, Jay L. Karp, Samuel Silverman, Mohar Kalra, Hollis Lehv, Faizan Jamil, Brian A. Smith
*Navstick: Making Video Games Blind-Accessible via the Ability to Look Around*
**ACM Symposium on User Interface Software and Technology (2021)**

## Employment

**Charles River Analytics**, Cambridge, MA — March 2021 - August 2021
*Software Engineer Intern*
Sensing, Perception, and Applied Robotics Division

**MathWorks**, Natick, MA — May 2019 - August 2019
*Engineering Development Group Intern*

## Open-Source Software

Monad
C++ library for homogenization of structured 2D and 3D grids. It provides solvers and material models for computing effective (homogenized) material tensors from heterogeneous microstructures.

rwmicro
C++ library for generating random-walk microstructures—binary patterns created by a periodic random walk.

gcs
Python library for generating 3D meshes of generalized cylindrical shells.

## Invited Talks

*Random-Walk Microstructures for Differentiable Topology Optimization* — July 20, 2026
SIGGRAPH Graphics4Science Workshop
Los Angeles, CA

*Data-Driven Nonlinear Deformation Design of 3D-Printable Shells* — October 24, 2025
Materials Day 2025
Boston University, Boston, MA

*Data-Driven Nonlinear Deformation Design of 3D-Printable Shells* — March 27, 2023
Transformative Robotics Lab
Northeastern University, Boston, MA

*Data-Driven Nonlinear Deformation Design of 3D-Printable Shells* — March 8, 2023
Computational Fabrication Group
University of Washington, Seattle, WA

## Teaching

**Boston University**, Boston, MA
Teaching Assistant
*CAS CS 581: Computational Fabrication* — Spring 2024, 2026
*CAS CS 320: Concepts of Programming Languages* — Fall 2025
*CAS CS 411: Software Engineering* — Summer 2023

**Columbia University**, New York, NY
Course Assistant
*COMS 4170: User Interface Design* — Fall 2020

## Awards

**Dean's Fellowship**, Boston University, Boston, MA — 2021-2022

## Press Coverage

**The Brink**, *3D Printing Robot Uses AI Machine Learning for US Army Research* — May 2024